

Demystified Carathéodory extension theorem

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Let μ be a premeasure on an algebra $\mathcal{A}$ of subsets of S , and let μ^* be the outer measure induced by μ :

$$\mu^*(T) = \inf \left\{ \sum_{n=1}^{\infty} \mu(A_n) : T \subseteq \bigcup_{n=1}^{\infty} A_n, A_n \in \mathcal{A} \right\}.$$

As usual, $\mu^*(A) = \mu(A)$ for $A \in \mathcal{A}$.

Suppose that we want to adjoin one set $E \subseteq S$. Write

$$\mathcal{A}[E] = \text{alg}(\mathcal{A} \cup \{E\})$$

for the algebra generated by $\mathcal{A}$ and E . The outer measure μ^* is already defined on every subset of S , so there is an obvious question: does its restriction to $\mathcal{A}[E]$ give an extension of the premeasure with which we started?

Definition 1. A set $E \subseteq S$ is *Carathéodory measurable* if the restriction

$$\mu^*|_{\mathcal{A}[E]}$$

is a premeasure extending μ .

Thus E is measurable precisely when it can be adjoined to the algebra without changing the outer measure already constructed.

The usefulness of the definition comes from alternative characterisations. We first record one elementary property of the original algebra.

Lemma 1. For every $A \in \mathcal{A}$ and every $T \subseteq S$,

$$\mu^*(T) = \mu^*(T \cap A) + \mu^*(T \cap A^c). \quad (1)$$

Proof. Subadditivity gives the inequality “ $\leq$ ”. For the reverse inequality, let

$$T \subseteq \bigcup_n B_n, \quad B_n \in \mathcal{A}.$$

Then

$$T \cap A \subseteq \bigcup_n (B_n \cap A), \quad T \cap A^c \subseteq \bigcup_n (B_n \cap A^c),$$

and hence

$$\begin{aligned}\mu^*(T \cap A) + \mu^*(T \cap A^c) &\leq \sum_n \mu(B_n \cap A) + \sum_n \mu(B_n \cap A^c) \\ &= \sum_n \mu(B_n).\end{aligned}$$

Taking the infimum over all such covers proves the reverse inequality. $\square$

Proposition 1 (Useful characterisations). *For $E \subseteq S$, the following are equivalent.*

1. E is Carathéodory measurable in the sense of the definition above.

2. For every $A \in \mathcal{A}$,

$$\mu^*(A) = \mu^*(A \cap E) + \mu^*(A \cap E^c). \quad (2)$$

3. For every $T \subseteq S$,

$$\mu^*(T) = \mu^*(T \cap E) + \mu^*(T \cap E^c). \quad (3)$$

Proof. If $\mu^*|_{\mathcal{A}[E]}$ is a premeasure, then for $A \in \mathcal{A}$ the disjoint union

$$A = (A \cap E) \cup (A \cap E^c)$$

gives (2). Thus (1) implies (2).

Suppose now that (2) holds, and let $T \subseteq S$. For every algebra cover

$$T \subseteq \bigcup_n A_n, \quad A_n \in \mathcal{A},$$

we have

$$\begin{aligned}\mu^*(T \cap E) + \mu^*(T \cap E^c) &\leq \sum_n \mu^*(A_n \cap E) + \sum_n \mu^*(A_n \cap E^c) \\ &= \sum_n \mu^*(A_n) = \sum_n \mu(A_n).\end{aligned}$$

Taking the infimum over all such covers gives

$$\mu^*(T \cap E) + \mu^*(T \cap E^c) \leq \mu^*(T).$$

The reverse inequality is subadditivity, so (3) follows.

It remains to show that (3) implies (1). Let $\mathcal{C}$ be the class of all $F \subseteq S$ satisfying

$$\mu^*(T) = \mu^*(T \cap F) + \mu^*(T \cap F^c) \quad \text{for every } T \subseteq S. \quad (4)$$

By Lemma 1, $\mathcal{A} \subseteq \mathcal{C}$. The class $\mathcal{C}$ is closed under complements. It is also closed under finite unions: if $F, G \in \mathcal{C}$, then

$$\begin{aligned}\mu^*(T) &= \mu^*(T \cap F) + \mu^*(T \cap F^c) \\ &= \mu^*(T \cap F) + \mu^*(T \cap F^c \cap G) + \mu^*(T \cap F^c \cap G^c) \\ &\geq \mu^*(T \cap (F \cup G)) + \mu^*(T \cap (F \cup G)^c),\end{aligned}$$

and subadditivity gives the reverse inequality. Hence $\mathcal{C}$ is an algebra.

If E satisfies (3), then $E \in \mathcal{C}$, and therefore

$$\mathcal{A}[E] \subseteq \mathcal{C}.$$

Let $B_n \in \mathcal{A}[E]$ be pairwise disjoint and suppose that $B = \bigcup_n B_n$ belongs to $\mathcal{A}[E]$. Since each $B_n \in \mathcal{C}$, repeated use of (4), with $T = B$, gives for every N

$$\mu^*(B) = \sum_{n=1}^N \mu^*(B_n) + \mu^*\left(B \setminus \bigcup_{n=1}^N B_n\right).$$

Thus

$$\mu^*(B) \geq \sum_{n=1}^{\infty} \mu^*(B_n),$$

while countable subadditivity gives the reverse inequality. Hence $\mu^*|_{\mathcal{A}[E]}$ is a premeasure. Since $\mu^* = \mu$ on $\mathcal{A}$, it extends μ . This is (1). $\square$

The point of the last characterisation is that measurability no longer refers to the algebra currently under consideration. If one tried to adjoin sets successively using only a condition on the current algebra, the algebra would change at every step. Condition (3) removes this bookkeeping: it is a condition on E and the fixed outer measure μ^* alone. We can therefore ask for all sets that are admissible in this sense at once. This is exactly where Carathéodory's theorem enters.

Theorem 1 (Carathéodory). *The Carathéodory measurable sets form a σ -algebra $\mathcal{M}$ containing $\mathcal{A}$, and $\mu^*|_{\mathcal{M}}$ is a complete measure extending μ .*

Proof. By Proposition 1, $\mathcal{M}$ is exactly the class $\mathcal{C}$ defined by (4). We already know that $\mathcal{C}$ is an algebra containing $\mathcal{A}$. It remains to prove closure under countable unions.

Let $E_n \in \mathcal{C}$ be pairwise disjoint and put

$$E = \bigcup_{n=1}^{\infty} E_n.$$

For every $T \subseteq S$, repeated use of (4) gives

$$\mu^*(T) = \sum_{n=1}^N \mu^*(T \cap E_n) + \mu^*\left(T \cap \left(\bigcup_{n=1}^N E_n\right)^c\right).$$

Since

$$T \cap E^c \subseteq T \cap \left(\bigcup_{n=1}^N E_n\right)^c,$$

we obtain

$$\mu^*(T) \geq \sum_{n=1}^N \mu^*(T \cap E_n) + \mu^*(T \cap E^c).$$

Letting $N \rightarrow \infty$ and using countable subadditivity on $T \cap E = \bigcup_n (T \cap E_n)$ gives

$$\mu^*(T) \geq \mu^*(T \cap E) + \mu^*(T \cap E^c).$$

The reverse inequality is subadditivity. Hence $E \in \mathcal{C}$.

For a general sequence $E_n \in \mathcal{C}$, replace it by the disjoint sequence

$$F_1 = E_1, \quad F_n = E_n \setminus \bigcup_{k < n} E_k.$$

Since $\mathcal{C}$ is an algebra, $F_n \in \mathcal{C}$, and the union is unchanged. Thus $\mathcal{C}$ is a σ -algebra.

Now let $E_n \in \mathcal{M}$ be pairwise disjoint and put $E = \bigcup_n E_n$. Since $E \in \mathcal{M}$, the same finite splitting argument with $T = E$ gives

$$\mu^*(E) \geq \sum_{n=1}^{\infty} \mu^*(E_n),$$

while countable subadditivity gives the reverse inequality. Therefore $\mu^*|_{\mathcal{M}}$ is a measure. It extends μ because $\mathcal{A} \subseteq \mathcal{M}$ and $\mu^* = \mu$ on $\mathcal{A}$.

Finally, suppose $N \in \mathcal{M}$ and $\mu^*(N) = 0$, and let $E \subseteq N$. Then $\mu^*(T \cap E) = 0$ for every $T \subseteq S$. Since

$$T = (T \cap E) \cup (T \cap E^c),$$

subadditivity and monotonicity give

$$\mu^*(T) = \mu^*(T \cap E^c) = \mu^*(T \cap E) + \mu^*(T \cap E^c).$$

Thus $E \in \mathcal{M}$, so the measure is complete. □